

‘FraudShield’

A Credit Card Fraud Detection and Transaction Monitoring System

A project report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

(Department of Computer Science & Engineering)

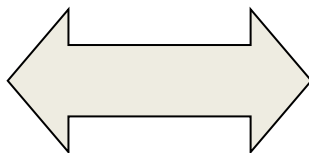
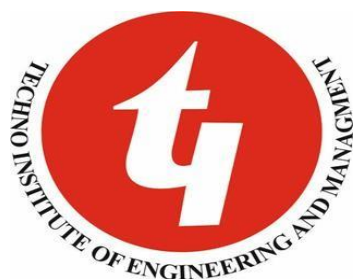
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DECLARATION OF ORIGINALITY AND COMPLIANCE

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We, hereby declare that the thesis contains a literature survey and original research work by the undersigned candidates and the project report entitled "**FraudShield:A Credit Card Fraud Detection and Transaction Monitoring System**" is submitted in partial fulfillment of the requirement for the award of the degree of **BACHELOR OF TECHNOLOGY** in **COMPUTER SCIENCE & ENGINEERING** .

We hereby declare that all information in this document has been obtained and presented by academic rules and ethical conduct.

We also declare that, as required by these rules and conduct, we have fully cited and referenced all material and results that are not original to this work.

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TITLE OF THE PROJECT

‘FraudShield’

A Credit Card Fraud Detection and Transaction Monitoring System

Abstract

The rapid growth of digital banking, online shopping, and cashless payment systems has significantly increased the use of credit and debit cards. Along with this growth, credit card fraud has also become one of the major challenges faced by banks, financial institutions, and customers. Fraudulent transactions can result in financial losses, identity theft, and reduced customer trust. Traditional fraud detection systems mainly rely on fixed rules, which often fail to identify new and evolving fraud patterns. Therefore, there is a need for an intelligent system that can detect suspicious transactions quickly and accurately while helping users monitor their banking activities.

The Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning is a web-based application developed to identify and monitor fraudulent credit card transactions. The system combines Machine Learning with a Rule-Based Security Engine to improve fraud detection accuracy. It uses a Random Forest Classifier model to analyze transaction details such as transaction amount, transaction hour, merchant category, account age, failed PIN attempts, transaction location, and the number of transactions made within the last hour. Based on these features, the model predicts the probability of fraud and generates a **risk score** that classifies transactions into Safe, Medium Risk, or High Risk/Fraudulent categories.

Apart from Machine Learning, the system also implements predefined banking security rules to enhance transaction safety. These rules verify account balance, daily transaction limits, maximum transaction amount limits, and failed PIN attempts before processing a transaction. If a transaction violates any of these rules or is identified as suspicious by the Machine Learning model, the system generates alerts, stores the transaction details, and sends email notifications to the registered user. The application also provides an interactive dashboard where users can monitor transaction statistics, visualize fraud trends through charts, access transaction history, review alerts, and generate PDF and Excel reports.

The system is developed using Python, Flask, SQLite, Scikit-learn, HTML, CSS, JavaScript, and Chart.js, providing a simple yet effective platform for fraud detection and transaction monitoring. The proposed solution demonstrates how Machine Learning and Rule-Based Security can work together to improve transaction security, reduce fraudulent activities, and assist users in monitoring their financial transactions. In the future, the system can be enhanced by integrating live banking APIs such as Plaid, implementing OTP verification, SMS notifications, and real-time transaction monitoring to make it more suitable for real-world banking environments.

1.Introduction

The advancement of digital technology has transformed the banking and financial sector by enabling secure online payments, internet banking, mobile banking, and digital wallet services. Today, millions of financial transactions are performed every day using credit cards and debit cards because they provide convenience, speed, and flexibility to users. However, the increasing use of digital payment systems has also led to a significant rise in fraudulent activities. Cybercriminals continuously develop new techniques such as stolen card usage, phishing attacks, card cloning, identity theft, and unauthorized transactions to exploit weaknesses in payment systems. As a result, protecting customers from financial fraud has become a major priority for banks and financial institutions.

Traditional fraud detection systems mainly rely on predefined rules, such as transaction amount limits or unusual transaction locations, to identify suspicious activities. Although these systems are effective in detecting simple fraud patterns, they often struggle to recognize newly emerging or complex fraudulent behaviors. Machine Learning has become an important solution for this problem because it can learn from historical transaction data and identify hidden patterns that may indicate fraudulent activity. By analyzing multiple transaction features simultaneously, Machine Learning models can provide more accurate and intelligent fraud detection than traditional rule-based methods alone.

The Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning is designed to provide an intelligent solution for identifying and monitoring suspicious credit card transactions. The proposed system combines a Random Forest Machine Learning model with a Rule-Based Security Engine to improve fraud detection performance. Users register by providing their banking details, including account balance, transaction limits, and card information. During a transaction, the system analyzes various parameters such as transaction amount, transaction time, merchant category, failed PIN attempts, account age, and recent transaction frequency. The Rule Engine first validates predefined banking rules, while the Machine Learning model calculates the fraud probability and generates a risk score to classify each transaction as Safe, Medium Risk, or High Risk/Fraudulent.

The application also includes several additional features such as user authentication, dashboard analytics, transaction history, fraud alerts, email notifications, PDF and Excel report generation, and graphical visualization of transaction statistics. These features allow users to monitor their financial activities and quickly identify suspicious transactions. By combining intelligent fraud detection with transaction monitoring, the system helps improve transaction security and reduces the possibility of financial loss.

This project demonstrates the practical application of Machine Learning in financial cybersecurity and provides a scalable foundation for future enhancements such as live banking API integration, real-time transaction monitoring, OTP-based verification, SMS notifications, and cloud deployment. It highlights how modern technologies can be used to develop smarter and more reliable banking security systems that can adapt to evolving fraud patterns while improving the overall safety of digital financial transactions.

2.Objectives

The primary objective of the Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning is to develop an intelligent and secure web-based application that can identify fraudulent credit card transactions and assist users in monitoring their financial activities. The system combines Machine Learning with predefined security rules to improve the accuracy of fraud detection and reduce financial losses caused by unauthorized transactions.

The specific objectives of the proposed system are as follows:

- To develop a secure user registration and authentication system that allows only authorized users to access the application.
- To detect fraudulent credit card transactions using the Random Forest Machine Learning algorithm by analyzing various transaction features.
- To implement a Rule-Based Security Engine that validates transactions based on predefined banking rules such as account balance, daily transaction limit, maximum transaction amount, and failed PIN attempts.
- To calculate a risk score for every transaction based on the fraud probability predicted by the Machine Learning model and classify transactions into Safe, Medium Risk, or High Risk/Fraud.
- To generate alerts and send email notifications whenever suspicious or fraudulent transactions are detected, enabling users to take immediate action.
- To maintain a complete transaction history that allows users to review previous transactions and monitor their financial activities.
- To provide an interactive dashboard with graphical charts, transaction statistics, and recent transaction details for better visualization and analysis.
- To generate transaction reports in PDF and Excel formats for documentation and future reference.
- To securely store user information, banking details, transaction records, fraud predictions, alerts, and reports using an SQLite database.
- To develop a scalable and user-friendly system that can be enhanced in the future by integrating live banking APIs, real-time transaction monitoring, OTP verification, SMS notifications, and cloud deployment.

The overall objective of this project is to provide an efficient, accurate, and user-friendly fraud detection system that enhances the security of digital financial transactions, minimizes financial losses, and improves user confidence in online banking and payment systems.

3.Problem Statement

The increasing use of digital banking, online shopping, and electronic payment systems has significantly improved the convenience of financial transactions. However, this rapid digital transformation has also led to a substantial increase in credit card fraud and unauthorized financial activities. Fraudsters continuously adopt advanced techniques such as stolen card usage, identity theft, phishing attacks, card cloning, and unauthorized online transactions, making fraud detection more challenging than ever before.

Traditional fraud detection systems mainly rely on predefined rules to identify suspicious transactions. Although these systems can detect basic fraud patterns, they are often unable to recognize new or evolving fraudulent activities. As fraud techniques become more sophisticated, fixed rule-based systems produce higher false positives and may fail to detect genuine fraudulent transactions. This increases financial losses for banks, businesses, and customers while reducing trust in digital payment systems.

Another major challenge is the lack of continuous transaction monitoring and timely notification mechanisms. Many existing systems either notify users after a fraudulent transaction has already been completed or fail to provide sufficient information for monitoring suspicious activities. This delay can result in unauthorized financial losses before preventive action can be taken.

To address these challenges, there is a need for an intelligent fraud detection system that combines Machine Learning with Rule-Based Security to improve fraud detection accuracy while providing transaction monitoring and timely alerts. The proposed Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning addresses this problem by analyzing multiple transaction parameters, calculating fraud probability, generating a risk score, and classifying transactions into Safe, Medium Risk, or High Risk categories. The system also maintains transaction history, generates dashboard analytics and reports, and sends email notifications for suspicious transactions, thereby enhancing transaction security and reducing the risk of financial fraud.

This project aims to provide a practical, scalable, and efficient solution that demonstrates how intelligent technologies can improve fraud detection and transaction monitoring while serving as a foundation for future integration with real-time banking APIs and advanced financial security mechanisms.

4.Literature Review

One of the studies, "Credit Card Fraud Detection in the Era of Disruptive Technologies: A Systematic Review," discussed different approaches such as traditional machine learning, deep learning, feature engineering, and techniques for handling imbalanced datasets. It concluded that combining advanced machine learning algorithms with proper data preprocessing improves fraud detection accuracy. However, the study mainly reviewed existing techniques and did not propose a practical fraud detection system for real-world implementation. [1]

Another study, "Credit Card Fraud Detection Using Machine Learning: A Survey," reviewed several machine learning algorithms including Logistic Regression, Random Forest, Hidden Markov Models, and Long Short-Term Memory (LSTM) networks for detecting fraudulent credit card transactions. The study emphasized that fraud detection models should continuously adapt because fraud patterns and customer behavior change over time. However, the research mainly focused on comparing algorithms and did not develop a complete web-based fraud detection application. [2]

The paper "How Artificial Intelligence and Machine Learning Research Impacts Payment Card Fraud Detection: A Survey and Industry Benchmark" examined the use of artificial intelligence and machine learning techniques in payment card fraud detection. The study compared different AI-based approaches with industry requirements and found that only a few techniques are suitable for large-scale banking environments. It recommended combining artificial intelligence with operational fraud prevention methods to improve detection performance. However, the paper focused mainly on industry analysis and did not implement a practical fraud detection system. [3]

Another research paper titled "Review of Machine Learning Approach on Credit Card Fraud Detection" compared several machine learning algorithms such as Random Forest, Artificial Neural Networks, Support Vector Machines, and K-Nearest Neighbors. The study concluded that the Random Forest algorithm provides better fraud detection accuracy because it reduces overfitting and improves classification performance. It also highlighted the importance of privacy and security in financial applications. However, the proposed study did not include transaction monitoring or reporting features. [4]

The paper "Credit Card Fraud Detection Using Machine Learning and Data Mining Techniques – A Literature Survey" reviewed both machine learning and data mining techniques used in fraud detection systems. The authors explained that fraudsters continuously change their attack methods, making adaptive algorithms essential for effective fraud detection. The study identified Random Forest, Decision Trees, Logistic Regression, and Neural Networks as the most commonly used techniques and suggested combining multiple algorithms to improve prediction accuracy. However, the research was limited to a literature survey and did not implement a complete fraud detection application. [5]

The study "A Systematic Review of Literature on Credit Card Cyber Fraud Detection Using Machine and Deep Learning" compared various machine learning and deep learning models, evaluated their performance, and discussed current research trends and future directions. The study recommended further research on semi-supervised and unsupervised learning methods because fraud datasets are highly imbalanced. However, it focused mainly on algorithm comparison rather than developing a complete fraud detection system. [6]

Another study, "Literature Review of Credit Card Fraud Detection with Machine Learning," presented a detailed review of fraud detection techniques using machine learning algorithms. It discussed the use of Genetic Algorithms combined with Random Forest, Decision Trees, and Artificial Neural Networks for improving fraud detection accuracy. The research concluded that hybrid machine learning models and proper feature selection

significantly enhance prediction performance. However, the study did not include real-time transaction monitoring or user interaction modules. [7]

The paper "Credit Card Fraud Detection Techniques: A Survey" compared several machine learning algorithms including Random Forest, Naïve Bayes, Logistic Regression, K-Nearest Neighbors, and Multilayer Perceptron. The study evaluated these algorithms based on fraud detection accuracy and computational efficiency. It concluded that Random Forest consistently performs better because of its ensemble learning capability and robustness against noisy data. However, the study provided limited discussion on implementing fraud detection systems in real-world banking applications. [8]

The research paper "A Review of Machine Learning Applications for Credit Card Fraud Detection with a Case Study" reviewed supervised learning techniques such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost. A case study demonstrated that ensemble learning methods provide better fraud detection performance than individual algorithms. The paper also discussed data preprocessing methods to handle imbalanced datasets. However, the study mainly focused on prediction accuracy and did not include complete transaction monitoring or system architecture. [9]

The study "**Machine Learning Methods for Credit Card Fraud Detection: A Survey**" provided an overview of recent developments in machine learning for fraud detection. It discussed traditional machine learning methods, deep learning techniques, graph-based models, feature engineering, and explainable artificial intelligence. The authors emphasized the importance of scalable and interpretable fraud detection models for real-world financial institutions. However, the study focused mainly on algorithmic improvements and did not present a complete web-based fraud management system. [10]

5. Proposed Methodology

5.1 Introduction

The proposed Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning is designed to identify fraudulent credit card transactions while providing users with a secure platform to monitor their financial activities. The system combines Machine Learning with a Rule-Based Security Engine to improve fraud detection accuracy and reduce unauthorized financial transactions.

Unlike traditional systems that depend only on predefined banking rules, the proposed system integrates a Random Forest Classifier, enabling it to learn patterns from historical transaction data and predict fraudulent activities more effectively. Along with fraud detection, the system also provides dashboard analytics, transaction history, fraud alerts, report generation, and email notifications.

The proposed methodology consists of several modules that work together to provide secure transaction monitoring and fraud detection.

5.2 Overall Working Methodology

The overall workflow of the system is illustrated below.

User Registration ---> User Login --> Dashboard --> Enter Transaction Details --> Rule Engine Validation -----> Machine Learning Model(Random Forest)--> Fraud Probability --> Risk Score Generation ----> Safe / Medium Risk / Fraud --> Store Transaction --> Dashboard Update --> Email Notification --> Reports & Alerts

5.3 User Registration Module

The first module of the system is the user registration process. Every new user must create an account before accessing the application.

During registration, the user provides:

- Full Name
- Email Address
- Password
- Phone Number
- Bank Name
- Card Number
- Account Number
- Current Account Balance
- Daily Transaction Limit
- Maximum Transaction Amount

The provided information is validated and stored securely in the SQLite database.

This information is later used by the Rule Engine to verify banking constraints during transaction processing.

5.4 Authentication Module

After successful registration, users log into the application using their registered email address and password.

The authentication module verifies user credentials before allowing access to the dashboard.

This module provides:

- Secure Login
- Session Management
- User Authentication
- Logout Functionality

Only authenticated users are allowed to perform transactions and access reports.

5.5 Transaction Processing Module

Once logged in, users can analyze transactions manually through the transaction form.

The transaction form collects the following information:

- Transaction Amount
- Transaction Hour
- Transaction Place
- Transactions in Last Hour
- Failed PIN Attempts
- Account Age (Days)
- Merchant Category

These transaction features are used as inputs for both the Rule Engine and the Machine Learning model.

5.6 Rule-Based Security Engine

Before Machine Learning prediction, the transaction is verified using predefined banking rules.

The Rule Engine performs the following validations:

Rule 1 :

Balance Verification If

Transaction Amount > Available Balance

Action:

Transaction Blocked

Rule 2:

Daily Transaction Count Verification ,If

Today's Transaction Count > User Daily Transaction Limit

Action:

- Transaction Blocked
- Email Notification Sent

Rule 3:

Maximum Transaction Amount Verification,If

Transaction Amount > User Maximum Transaction Amount

Action:

- Transaction Blocked
- Email Notification Sent

Rule 4:

Failed PIN Attempt Verification

If

Failed PIN Attempts exceed the threshold

Action:

- Mark Transaction as High Risk
- These rules provide an additional layer of security before Machine Learning analysis.

5.7 Machine Learning Module

If the transaction satisfies the Rule Engine validations, it is forwarded to the Machine Learning model.

The project uses the Random Forest Classifier because it offers:

- High Accuracy
- Better Generalization
- Reduced Overfitting
- Robust Classification
- Fast Prediction

The model analyzes all transaction features and predicts the probability of fraud.

The prediction process consists of:

Transaction Features

↓

Random Forest Model

↓

Fraud Probability

↓

Risk Score

↓

Transaction Classification

5.8 Risk Score Generation

The fraud probability generated by the Machine Learning model is converted into a percentage.

$\text{Risk Score} = \text{Fraud Probability} \times 100$

The generated risk score determines the transaction category.

Risk Score	Status
0–40%	Safe
41–80%	Medium Risk
81–100%	High Risk/Fraud

The Risk Score provides users with an easy way to understand the fraud probability associated with each transaction.

5.9 Alert Generation

Based on the calculated Risk Score, the system automatically generates alerts.

Safe:

- Transaction Approved
- Stored in Database
- No Email Sent

Medium Risk:

- Transaction Stored
- Email Notification Sent

High Risk:

- Transaction Marked as Fraud
- Alert Generated
- Email Notification Sent

5.10 Dashboard Module

The dashboard provides a complete overview of user transactions.

The dashboard displays:

- Current Balance
- Total Transactions
- Safe Transactions
- Fraud Transactions
- Blocked Transactions
- Pie Chart
- Bar Chart
- Recent Transactions

The graphical representation helps users understand transaction statistics efficiently.

5.11 Reports Module

The reporting module enables users to generate transaction reports.

Features include:

- Date Filter
- Time Filter
- PDF Report Generation
- Excel Report Generation
- Transaction Summary

The reports help users review their financial activities over a selected period.

5.12 Email Notification Module

Whenever a suspicious transaction is detected, the system automatically sends an email notification using Gmail SMTP.

Email notifications are sent for:

- Medium Risk Transactions
- High Risk Transactions
- Daily Transaction Limit Exceeded
- Maximum Transaction Amount Exceeded

This ensures that users are informed immediately whenever suspicious activities occur.

5.13 Database Module

SQLite is used to store all project information.

The database stores:

- User Information
- Bank Details
- Transaction History
- Fraud Alerts
- Risk Scores
- Generated Reports

The database ensures data consistency and enables future transaction analysis.

6. Project Analysis

6.1 Introduction

Project analysis is an important phase in software development that involves studying the requirements, functionalities, and overall working of the proposed system before implementation. It helps identify user requirements, system behavior, input and output processing, and the interaction between different modules.

The proposed Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning has been analyzed to ensure that it provides secure transaction monitoring, intelligent fraud detection, user-friendly interaction, and efficient report generation. The system integrates Machine Learning with predefined banking security rules to improve fraud detection accuracy while maintaining transaction records and notifying users about suspicious activities.

6.2 Existing System Analysis

Most traditional banking fraud detection systems rely mainly on predefined security rules.

These systems generally perform checks such as:

- Transaction amount limit
- Daily transaction limit

Although these systems are useful, they have several limitations.

Limitations of Existing System:

- Unable to identify new fraud patterns.
- High false-positive rate.
- Limited adaptability.
- No intelligent learning capability.
- Lack of graphical transaction monitoring.
- Limited report generation.
- Delayed fraud notifications.
- Difficult to detect sophisticated fraud.

Because of these limitations, there is a need for an intelligent fraud detection system.

6.3 Proposed System Analysis

The proposed system overcomes the limitations of traditional systems by integrating Machine Learning with Rule-Based Security.

The system consists of multiple integrated modules including:

- User Authentication
- Transaction Monitoring
- Rule Engine
- Machine Learning Prediction
- Risk Score Calculation
- Dashboard
- Alerts
- Reports
- Email Notification

The combination of Rule-Based Security and Machine Learning significantly improves fraud detection efficiency.

6.4 Functional Requirements

The proposed system performs the following functions.

Authentication:User Registration ,Secure Login ,Session Management, Logout

Transaction Processing :Enter transaction details ,Validate transaction ,Perform fraud prediction ,Calculate risk score

Dashboard :Display current balance ,Show fraud statistics ,Display pie chart ,Display bar chart,Show recent transactions

Alerts :Store fraud alerts, Display fraud notifications ,Display blocked transactions

Reports :Generate PDF report ,Generate Excel report ,Filter by date

Email Notification :Notify users for suspicious transactions ,Notify users for blocked transactions

6.5 Non-Functional Requirements

The system should satisfy the following quality requirements.

Performance :The system should process transactions quickly.

Reliability :The application should provide consistent fraud prediction.

Security : Only authenticated users can access the application. Passwords are securely stored.

Usability :The interface should be simple and easy to use.

Scalability:The application should support future integration with banking APIs and cloud deployment.

6.6 Input Analysis

The system accepts the following inputs.

Registration Inputs

- Full Name
- Email
- Password
- Phone Number
- Bank Name
- Card Number
- Account Number
- Balance
- Daily Limit
- Maximum Amount

Transaction Inputs:

- Transaction Amount
- Transaction Hour
- Merchant Category
- Transaction Place
- Transactions Last Hour
- Failed PIN Attempts
- Account Age Days

These inputs are analyzed by both the Rule Engine and the Machine Learning model.

6.7 Output Analysis

The system generates the following outputs.

Fraud Prediction:

- Safe
- Medium Risk
- High Risk/Fraud

Dashboard:

Displays

- Current Balance
- Fraud Count
- Safe Transactions
- Pie Chart
- Bar Chart

Alerts:

Displays

- Fraud Alerts
- Blocked Transactions
- Alert Reason

Reports:

Generates

- PDF
- Excel

Email : Sends fraud notification to the registered email.

6.8 Feasibility Analysis

Technical Feasibility:

The system is technically feasible because it uses widely available technologies including Python, Flask, SQLite, and Scikit-learn.

Economic Feasibility:

The project requires only open-source software. Therefore, development cost is very low.

Operational Feasibility:

The application is user-friendly and can be operated without technical expertise.

7.Project Category

7.1 Introduction

The **Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning** is categorized as a **Web-Based Financial Security Application** that combines **Machine Learning**, **Rule-Based Security**, and **Transaction Monitoring** to detect fraudulent credit card transactions. The application is designed to improve banking transaction security by analyzing transaction data, predicting fraud probability, generating alerts, and providing graphical transaction analysis.

The project falls under multiple software development categories because it integrates web technologies, machine learning, database management, and cybersecurity concepts into a single application.

7.2 Project Classification

The proposed project belongs to the following categories:

Web-Based Application:

The system is developed as a web application using Flask and can be accessed through a web browser. Users can register, log in, perform transaction analysis, monitor fraud alerts, and generate reports through an interactive interface.

Machine Learning Application:

The project uses a **Random Forest Classifier** to predict whether a transaction is safe or fraudulent. The model analyzes multiple transaction features and calculates the fraud probability and risk score.

Financial Security System:

The application focuses on improving the security of digital banking and credit card transactions by detecting suspicious activities and preventing unauthorized transactions.

Decision Support System:

The application assists users by providing fraud predictions, risk scores, transaction history, alerts, and graphical analysis, helping them make informed decisions regarding transaction security.

Database-Driven Application:

The project stores user information, banking details, transaction records, fraud alerts, and reports in an SQLite database for secure storage and retrieval.

7.3 Application Area

The proposed system can be applied in:

- Banking Institutions
- Credit Card Companies
- Financial Organizations
- Online Payment Systems

8.Tools and Technology Used

8.1 Introduction

The proposed system is developed using modern web technologies, machine learning libraries, and database management tools. These technologies provide efficient development, secure data management, and accurate fraud prediction.

8.2 Frontend Technology

HTML:

HTML is used to design the user interface of the application. It creates web pages such as Login, Registration, Dashboard, Transaction Analysis, Alerts, History, and Reports.

Advantages:

- Simple
- Lightweight
- Platform Independent
- Easy Integration

8.3 Backend Technology

Python:

Python is the primary programming language used for application development.

Advantages:

- Easy to Learn, Secure
- Large Library Support

Flask:

Flask is a lightweight Python web framework used for developing the application.

Responsibilities:

- URL Routing
- User Authentication
- Database Connectivity
- Machine Learning Integration
- Session Management

Advantages:

- Lightweight , Easy API Integration
- Fast Development

8.4 Database

SQLite:

SQLite stores:

- User Details
- Transaction History
- Fraud Alerts
- Banking Information
- Reports

Advantages:

- Lightweight
- Serverless
- Easy Deployment
- Fast Query Execution

8.5 Machine Learning

Scikit-learn:

Used for:

- Training Machine Learning Models
- Fraud Prediction
- Model Evaluation

Random Forest Classifier:

Purpose:

Detect fraudulent transactions.

Advantages:

- High Accuracy
- Less Overfitting
- Handles Large Datasets
- Robust Classification

Joblib:

Used to save and load the trained Machine Learning model.

8.6 Data Processing Libraries

Pandas:

Used for:

- Data Cleaning
- Dataset Handling
- Feature Selection
- Data Manipulation

NumPy:

Used for:

- Numerical Computation
- Matrix Operations
- Feature Conversion

8.7 Data Visualization

Chart.js:

Used to generate:

- Pie Charts
- Bar Charts

The charts help users analyze transaction statistics visually.

8.8 Report Generation

ReportLab:

Used to generate PDF transaction reports.

OpenPyXL:

Used to generate Excel reports.

8.9 Email Notification

Gmail SMTP:

Used to send email notifications for:

- Medium Risk Transactions
- High Risk Transactions
- Transaction Limit Exceeded
- Maximum Amount Exceeded

8.10 Development Environment

Visual Studio Code:

Used for:

- Coding
- Debugging
- Project Management

9. Software Requirement Analysis

9.1 Introduction

Software Requirement Analysis (SRA) identifies the functional and non-functional requirements necessary for developing the proposed system. Proper requirement analysis ensures that the system meets user expectations while maintaining security, reliability, and performance.

9.2 Functional Requirements

The system must provide the following functionalities:

User Module:

- User Registration , Login
- Logout
- Session Management

Transaction Module:

- Manual Transaction Analysis
- Fraud Detection
- Risk Score Calculation

Dashboard Module:

- View Transaction Statistics
- View Pie Charts , Bar Charts
- View Recent Transactions

Alerts Module:

- Display Fraud Alerts
- Display Medium Risk Alerts
- Display Blocked Transactions

Reports Module:

- Generate PDF Reports
- Generate Excel Reports
- Filter Reports

Email Module:

- Send Fraud Alerts
- Send Transaction Notifications

9.3 Non-Functional Requirements

Performance:

The system should analyze transactions quickly with minimum delay.

Reliability:

The system should provide consistent and accurate fraud prediction.

Security:

- User Authentication
- Password Protection
- Secure Database Storage
- Email Verification

Scalability:

The system should support future integration with:

- Banking APIs
- OTP Authentication
- Cloud Deployment

Availability:

The application should remain available whenever users need to monitor transactions.

9.4 Hardware Requirements

Minimum Requirements:

- Processor: Intel Core i3 or above
- RAM: 4 GB
- Hard Disk: 20 GB Free Space
- Internet Connection
- Keyboard & Mouse

Recommended Requirements:

- Intel Core i5/i7
- 8 GB RAM
- SSD Storage

9.5 Software Requirements

Operating System:

- Windows 10/11
- Linux
- macOS

Programming Language:

- Python 3.x

Framework:

- Flask

Database:

- SQLite3

Machine Learning:

- Scikit-learn

Libraries:

- Pandas
- NumPy
- Joblib
- ReportLab
- OpenPyXL

Frontend:

- HTML
- CSS
- JavaScript
- Bootstrap
- Chart.js

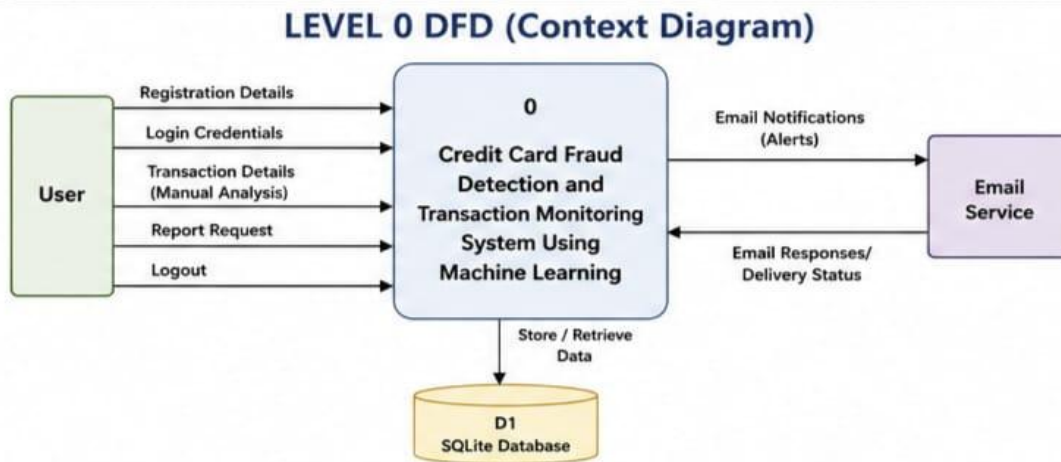
Development Tool:

- Visual Studio Code

10. Architectural Diagram

10.1 DFD

10.2.1 Level 0 DFD :



The Level 0 Data Flow Diagram (DFD), also known as the Context Diagram, provides a high-level overview of the Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning. It illustrates the interaction between the external entities and the system, showing how data flows into and out of the application. At this level, the entire application is represented as a single process without showing internal modules.

The process begins when the User interacts with the system by providing various inputs such as registration details, login credentials, transaction details for manual analysis, report requests, and logout requests. These inputs are received and processed by the Credit Card Fraud Detection and Transaction Monitoring System.

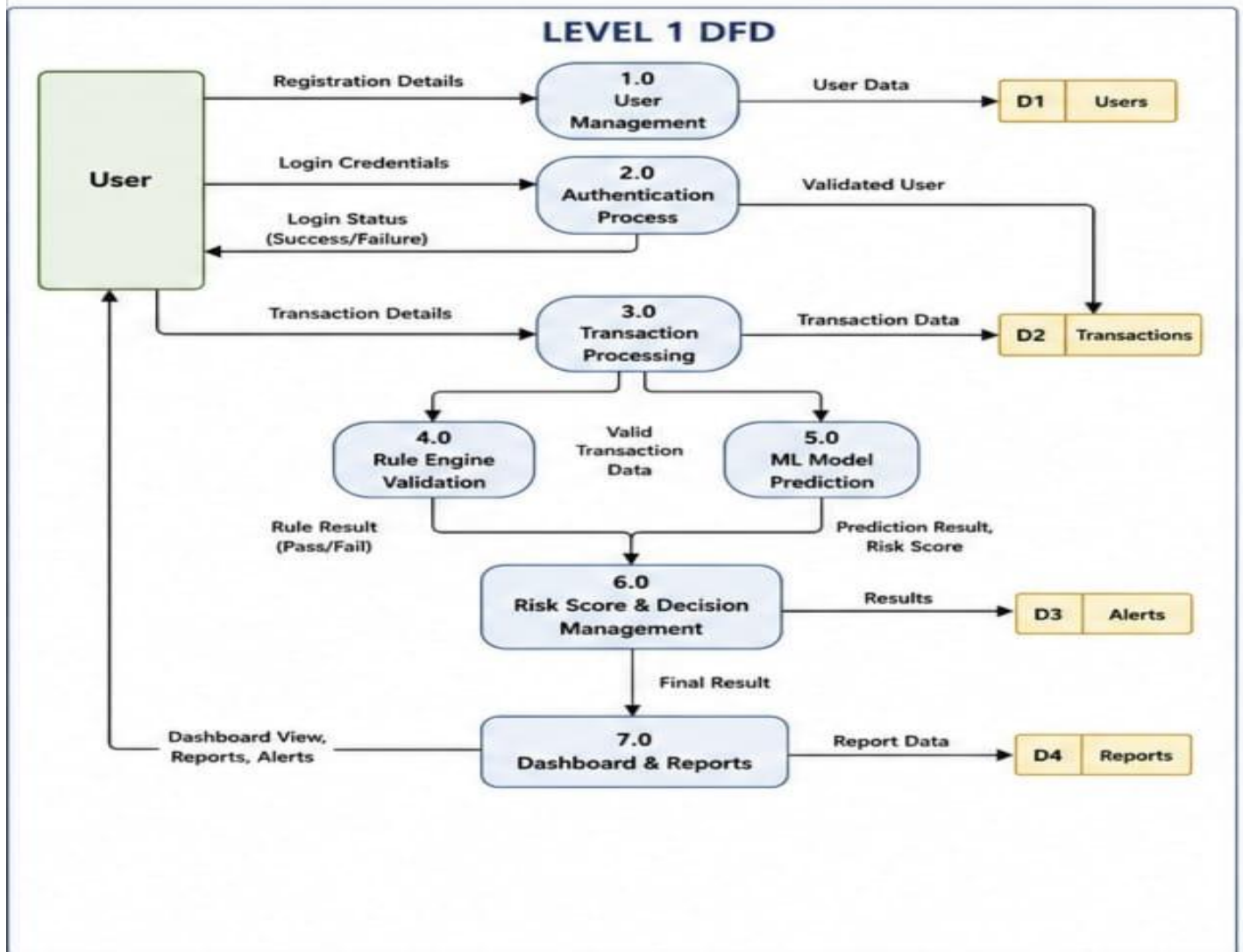
The system performs user authentication, validates transactions using predefined banking rules, analyzes transaction details using the Random Forest Machine Learning model, calculates the fraud probability and risk score, and determines whether the transaction is Safe, Medium Risk, or High Risk/Fraud.

All user information, transaction records, prediction results, alerts, and report details are stored and retrieved from the SQLite database (D1) whenever required. This database acts as the central repository for maintaining all application data.

If a transaction is identified as Medium Risk, High Risk, or violates any predefined banking rule, the system communicates with the Email Service to send email notifications to the registered user. The Email Service then returns the email delivery status or response back to the system, confirming whether the notification was successfully delivered.

Thus, the Level 0 DFD demonstrates the overall interaction between the User, the Fraud Detection System, the SQLite Database, and the Email Service, providing a simplified view of the complete system before examining the detailed internal processes in the higher-level DFDs.

10.2.2 Level 1 DFD :



The Level 1 Data Flow Diagram (DFD) provides a detailed view of the Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning by breaking the main process into its major functional modules. It illustrates how data flows between the user, system processes, and data stores during user registration, authentication, transaction processing, fraud detection, and report generation.

The process begins with the User, who interacts with the system by registering, logging in, and submitting transaction details. During User Management (1.0), the registration details are collected and stored in the Users database (D1).

Next, the Authentication Process (2.0) verifies the user's login credentials. If the credentials are valid, the user is successfully authenticated and allowed to access the system; otherwise, a login failure message is returned.

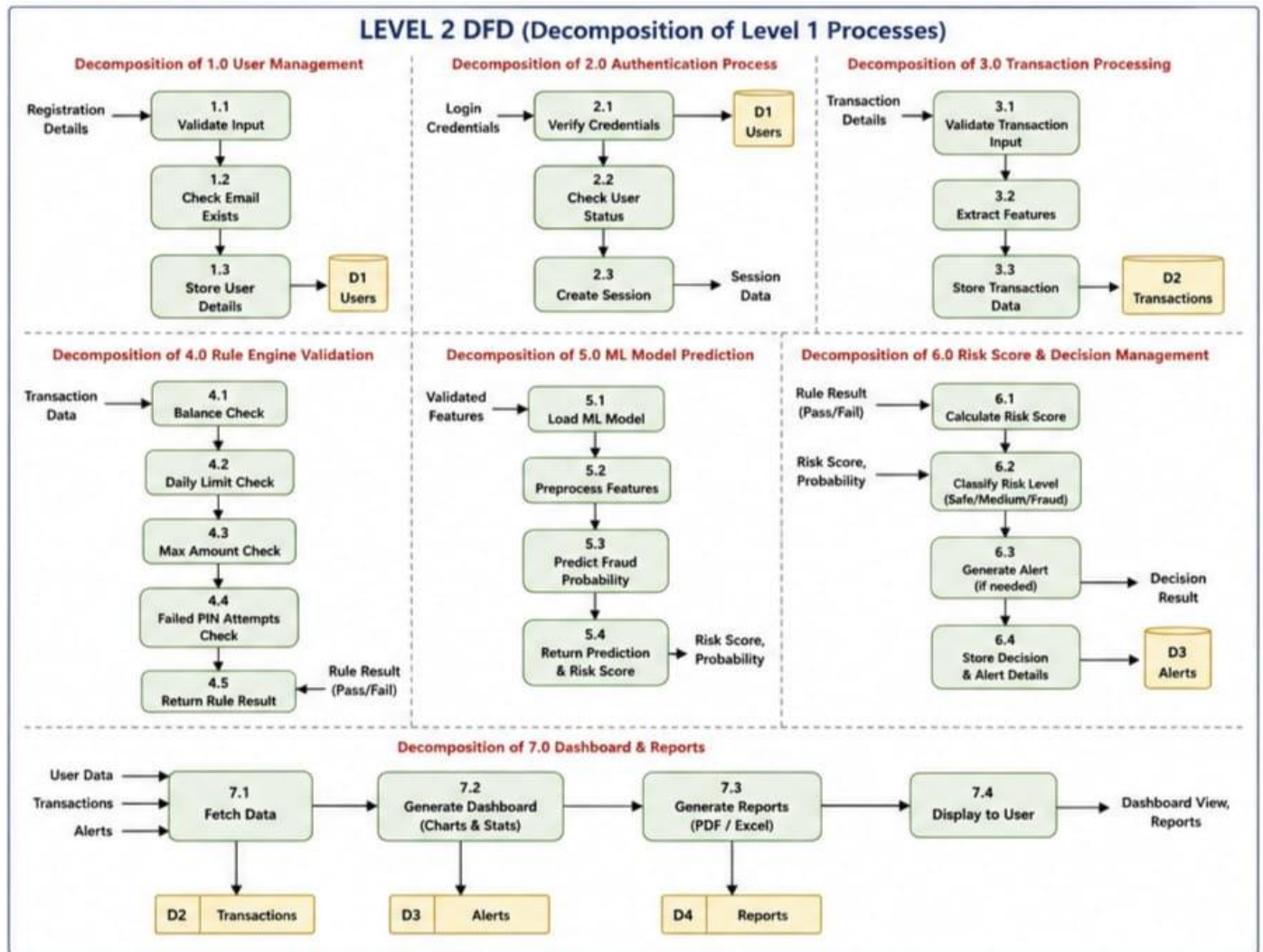
After successful authentication, the user enters transaction information through the Transaction Processing (3.0) module. The transaction details are stored in the Transactions database (D2) and simultaneously forwarded to the Rule Engine Validation (4.0) and Machine Learning Model Prediction (5.0) modules.

The Rule Engine checks predefined banking rules such as account balance, daily transaction limit, maximum transaction amount, and failed PIN attempts. At the same time, the Random Forest Machine Learning model analyzes the transaction and predicts the fraud probability along with the corresponding risk score.

The outputs from both modules are passed to the Risk Score and Decision Management (6.0) module, where the final transaction status is determined as Safe, Medium Risk, or High Risk/Fraud. If a suspicious or fraudulent transaction is detected, alert information is stored in the Alerts database (D3).

Finally, the processed transaction results are sent to the Dashboard and Reports (7.0) module. This module updates the dashboard with transaction statistics, fraud alerts, charts, and recent transactions while also generating PDF and Excel reports. The generated reports are stored in the Reports database (D4) and are made available to the user through the application.

10.2.3 Level 2 DFD:



The Level 2 Data Flow Diagram (DFD) provides a detailed decomposition of the major processes shown in the Level 1 DFD. It breaks each module into smaller sub-processes to explain how the Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning performs user management, authentication, transaction processing, rule validation, machine learning prediction, risk score calculation, dashboard generation, and report management.

User Management:

The User Management module begins when a new user submits registration details. The system first validates the input, checks whether the email already exists, and then stores the user's information in the Users (D1) database. This ensures that only valid and unique user accounts are created.

Authentication Process:

During authentication, the system verifies the user's login credentials using the information stored in the Users database. If the credentials are valid, the system checks the user status and creates a login session, allowing the user to access the application securely.

Transaction Processing:

After login, the user enters transaction details. The system validates the transaction input, extracts the required features such as amount, transaction time, merchant category, location, and failed PIN attempts, and stores the transaction information in the Transactions (D2) database for further processing.

Rule Engine Validation:

The transaction data is passed to the Rule Engine, where predefined banking rules are verified. The system performs balance verification, daily transaction limit check, maximum transaction amount check, and failed PIN attempts check. After completing these validations, the Rule Engine returns a Pass or Fail result for the transaction.

Machine Learning Model Prediction:

If the transaction passes the Rule Engine validation, the validated transaction features are processed by the Random Forest Machine Learning model. The system loads the trained model, preprocesses the transaction features, predicts the fraud probability, and returns both the fraud probability and the calculated risk score.

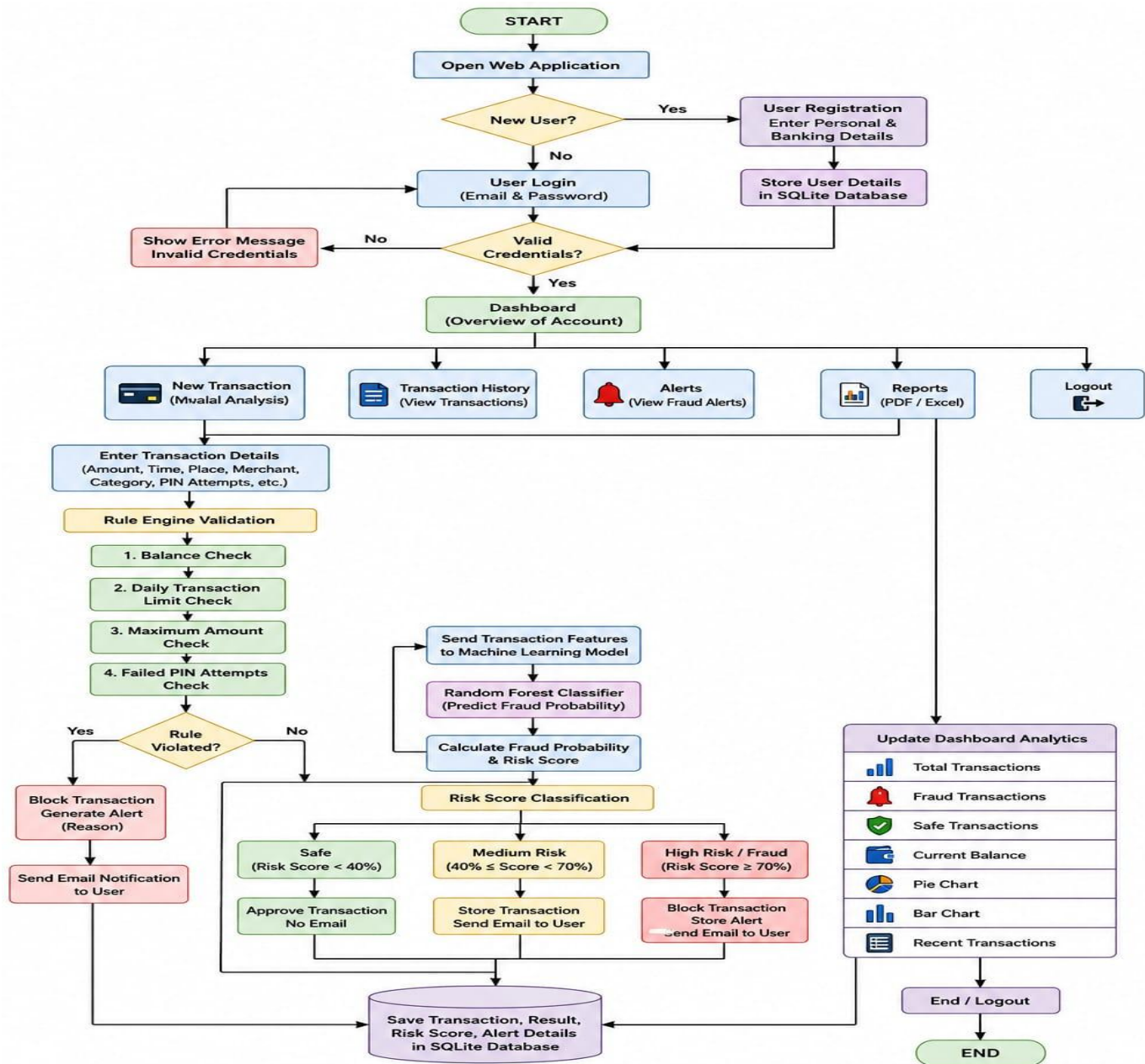
Risk Score and Decision Management:

The system receives the Rule Engine result along with the Machine Learning prediction. It calculates the final risk score, classifies the transaction as Safe, Medium Risk, or High Risk/Fraud, generates an alert if required, and stores the final decision and alert information in the Alerts (D3) database.

Dashboard and Reports:

Finally, the system retrieves user, transaction, and alert data to generate the dashboard and reports. It prepares graphical statistics such as charts and summaries, generates PDF and Excel reports, stores them in the Reports (D4) database, and displays the dashboard, transaction history, alerts, and reports to the user.

10.2 Flowchart



The flowchart illustrates the complete workflow of the Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning. It explains the sequence of operations from user registration and login to transaction analysis, fraud detection, report generation, and dashboard updates. The system combines Rule-Based Security with a Random Forest Machine Learning model to detect fraudulent transactions and provide secure transaction monitoring.

The process begins when the user opens the web application. If the user is new, they complete the registration process by entering personal and banking details, which are stored in the SQLite database. Existing users can directly log in using their registered email and password. The system validates the login credentials before granting access to the dashboard. If the credentials are invalid, an error message is displayed.

After successful login, the user reaches the dashboard, where they can perform various operations such as initiating a new transaction, viewing transaction history, checking fraud alerts, generating PDF or Excel reports, and logging out.

When the user selects New Transaction, they enter transaction details including the transaction amount, transaction time, merchant category, transaction location, account age, and failed PIN attempts. The transaction is first processed by the Rule Engine, which verifies predefined banking rules such as account balance, daily transaction limit, maximum transaction amount, and failed PIN attempts.

If any rule is violated, the transaction is immediately blocked. The system generates an alert, records the reason, sends an email notification to the registered user, and stores all details in the SQLite database.

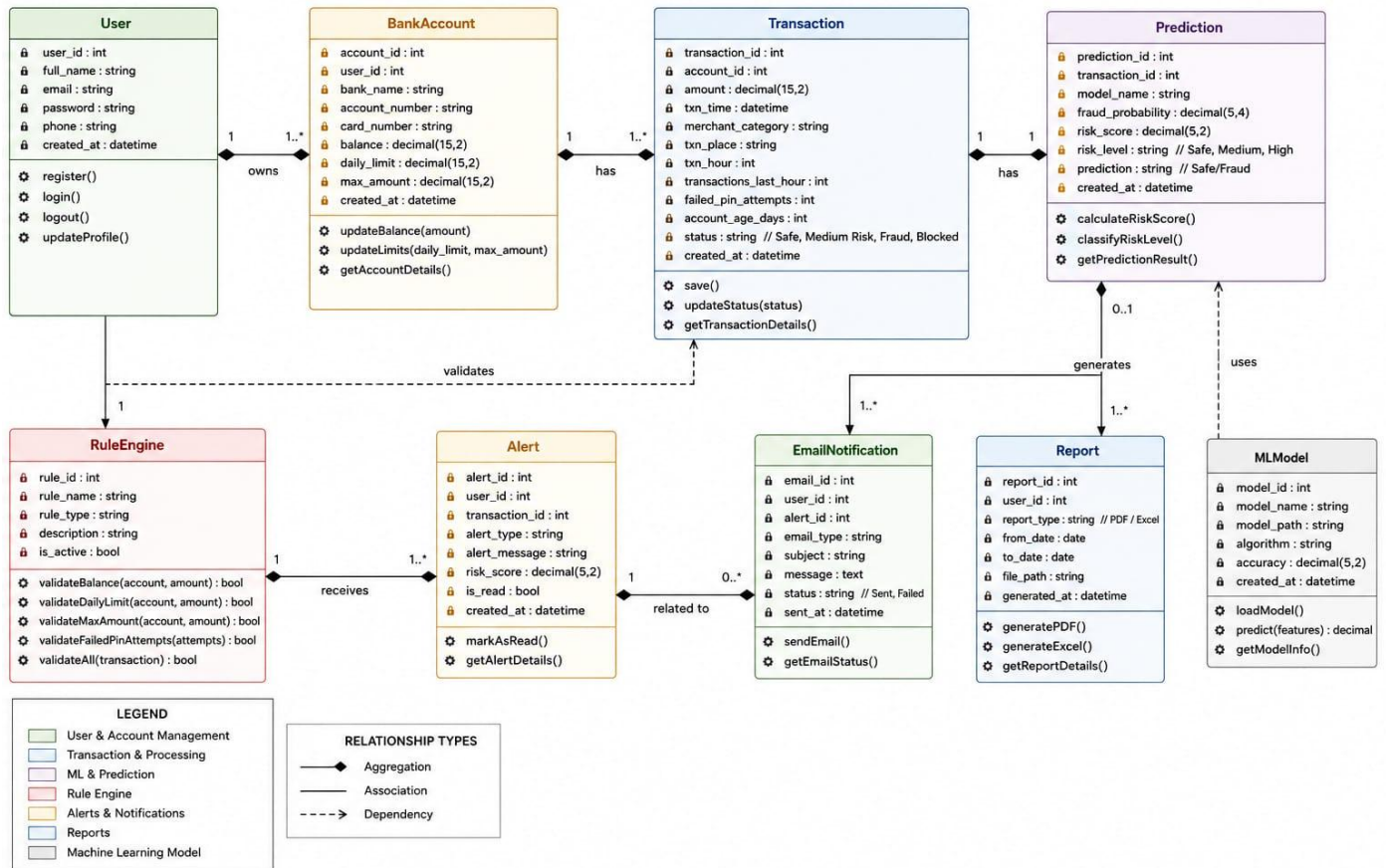
If the transaction passes all Rule Engine validations, the transaction features are forwarded to the Random Forest Machine Learning model. The model predicts the fraud probability, which is then converted into a risk score. Based on the calculated risk score, the transaction is classified into one of three categories:

- **Safe (Risk Score < 40%)** – The transaction is approved without sending an email notification.
- **Medium Risk ($40\% \leq \text{Risk Score} < 70\%$)** – The transaction is stored for monitoring, and an email notification is sent to the user.
- **High Risk/Fraud (Risk Score $\geq 70\%$)** – The transaction is blocked, an alert is generated, and an email notification is sent immediately.

Finally, all transaction details, prediction results, risk scores, alerts, and transaction status are stored in the SQLite database. The dashboard is automatically updated to display total transactions, fraud transactions, safe transactions, current account balance, pie charts, bar charts, and recent transaction history. The user can also generate reports or log out, completing the transaction process.

10.3 Class Diagram

CLASS DIAGRAM – CREDIT CARD FRAUD DETECTION AND TRANSACTION MONITORING SYSTEM



The Class Diagram represents the static structure of the Credit Card Fraud Detection and Transaction Monitoring System. It shows the main classes, their attributes, methods, and the relationships between them. Each class represents a major module of the system, and together they support fraud detection, transaction monitoring, alert generation, report creation, and email notifications.

User Class:

The User class stores all user-related information.

Attributes:

- User ID
- Full Name
- Email
- Password
- Phone Number
- Created Date

Methods:

- Register()
- Login()
- Logout()

- Update Profile()

A user can own one or more bank accounts and interact with all features of the system.

BankAccount Class:

The BankAccount class stores the user's banking information.

Attributes:

- Account ID
- User ID
- Bank Name
- Account Number
- Card Number
- Current Balance
- Daily Transaction Limit
- Maximum Transaction Amount

Methods:

- Update Balance()
- Update Limits()
- Get Account Details()

Each bank account is associated with one user and can have multiple transactions.

Transaction Class:

The Transaction class stores all transaction details entered by the user.

Attributes:

- Transaction ID
- Account ID
- Transaction Amount
- Transaction Time
- Merchant Category
- Transaction Place
- Transaction Hour
- Transactions Last Hour
- Failed PIN Attempts
- Account Age
- Transaction Status

Methods:

- Save()
- Update Status()

- Get Transaction Details()

Every transaction is validated, analyzed, and stored in the database.

Prediction Class:

The Prediction class stores the fraud detection results generated by the Machine Learning model.

Attributes:

- Prediction ID
- Transaction ID
- Model Name
- Fraud Probability
- Risk Score
- Risk Level
- Prediction Result
- Created Date

Methods:

- Calculate Risk Score()
- Classify Risk Level()
- Get Prediction Result()

It converts the fraud probability into a risk score and classifies the transaction as **Safe**, **Medium Risk**, or **High Risk**.

RuleEngine Class:

The RuleEngine class validates transactions using predefined banking rules before Machine Learning prediction.

Methods:

- Validate Balance()
- Validate Daily Limit()
- Validate Maximum Amount()
- Validate Failed PIN Attempts()
- Validate All Rules()

The Rule Engine blocks invalid transactions and forwards valid transactions for Machine Learning analysis.

Alert Class:

The Alert class stores alerts generated for suspicious or fraudulent transactions.

Attributes:

- Alert ID

- User ID
- Transaction ID
- Alert Type
- Alert Message
- Risk Score
- Read Status
- Created Date

Methods:

- Mark As Read()
- Get Alert Details()

Alerts are generated whenever a transaction is classified as Medium Risk or High Risk.

EmailNotification Class:

The EmailNotification class manages email alerts sent to users.

Attributes:

- Email ID
- User ID
- Alert ID
- Email Type
- Subject
- Message
- Email Status
- Sent Date

Methods:

- Send Email()
- Get Email Status()

It sends notification emails whenever suspicious transactions are detected.

Report Class:

The Report class generates downloadable transaction reports.

Attributes:

- Report ID
- User ID
- Report Type
- From Date
- To Date
- File Path

- Generated Date

Methods:

- Generate PDF()
- Generate Excel()
- Get Report Details()

Users can generate reports in PDF or Excel format.

MLModel Class:

The MLModel class represents the trained **Random Forest Machine Learning model** used for fraud detection.

Attributes:

- Model ID
- Model Name
- Model Path
- Algorithm
- Accuracy
- Created Date

Methods:

- Load Model()
- Predict()
- Get Model Information()

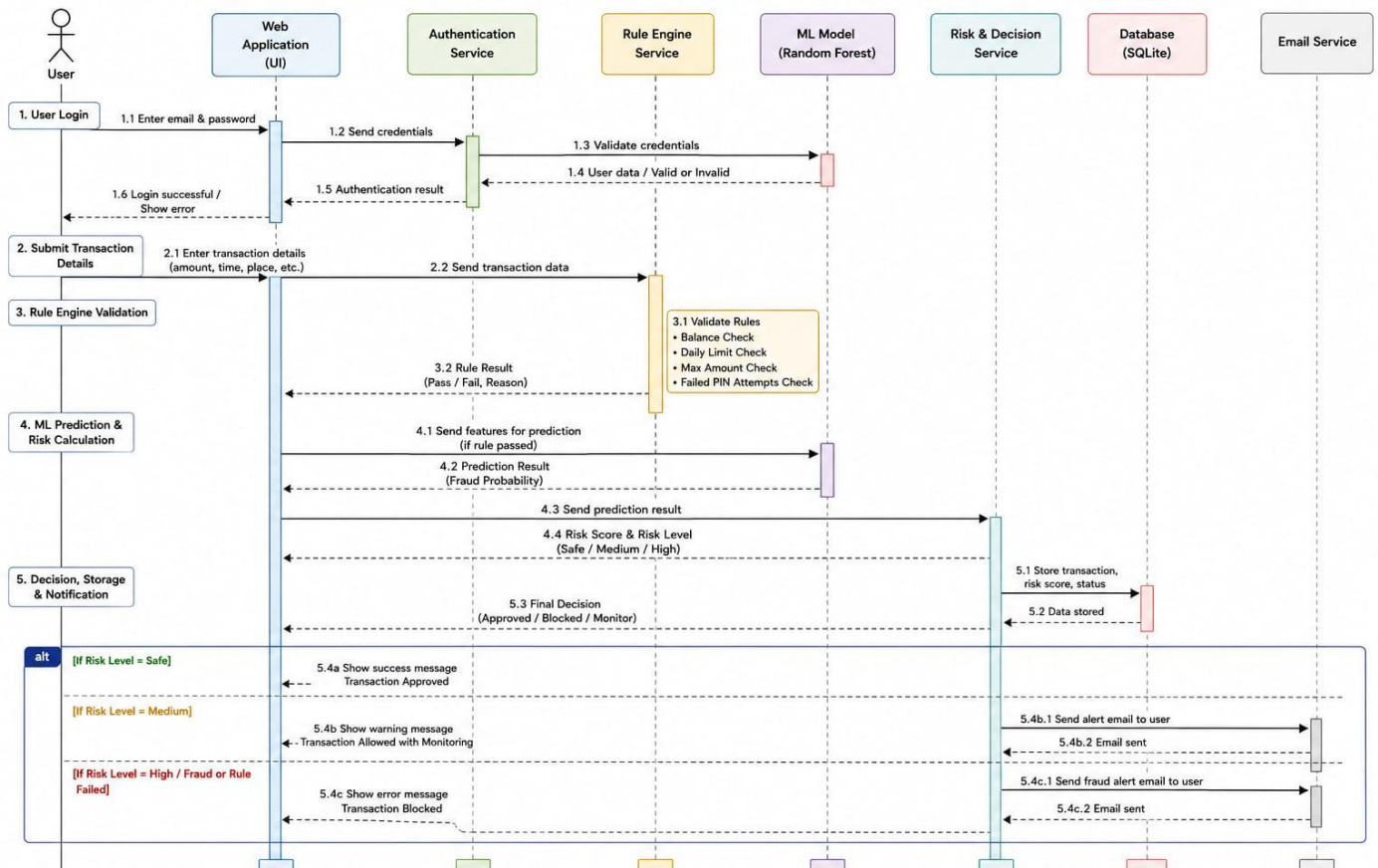
This class predicts fraud probability based on transaction features and returns the prediction to the system.

Relationships Between Classes

- **User → BankAccount:** One user can own one or more bank accounts.
- **BankAccount → Transaction:** One bank account can have multiple transactions.
- **Transaction → Prediction:** Each transaction has one prediction result.
- **RuleEngine → Transaction:** The Rule Engine validates every transaction before Machine Learning prediction.
- **Prediction → MLModel:** The Prediction class uses the trained Random Forest model to detect fraud.
- **RuleEngine → Alert:** Rule violations generate alerts.
- **Alert → EmailNotification:** Alerts trigger email notifications to the user.
- **Prediction → Report:** Prediction results are included in generated reports.

10.4 Sequence Diagram

SEQUENCE DIAGRAM – CREDIT CARD FRAUD DETECTION AND TRANSACTION MONITORING SYSTEM



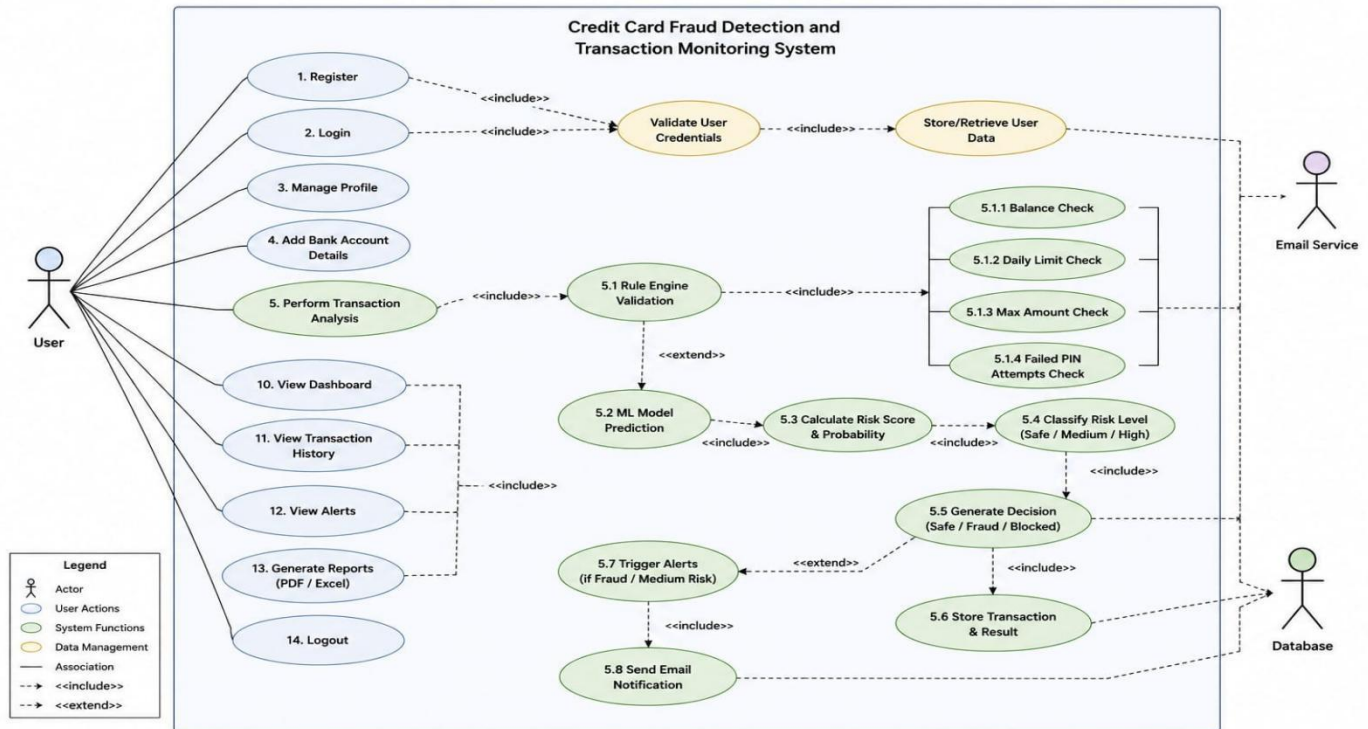
The sequence diagram illustrates the step-by-step interaction between the user and the various modules of the Credit Card Fraud Detection and Transaction Monitoring System. The process starts when the user logs into the application using valid credentials. After successful authentication, the user enters the transaction details such as transaction amount, transaction time, merchant category, transaction location, failed PIN attempts, and other required information.

The transaction is first processed by the Rule Engine, which verifies predefined banking rules including account balance, daily transaction limit, maximum transaction amount, and failed PIN attempts. If all the rules are satisfied, the transaction details are forwarded to the Random Forest Machine Learning model, which predicts the fraud probability and calculates the corresponding risk score.

Based on the calculated risk score, the Risk Decision Service classifies the transaction as Safe, Medium Risk, or High Risk/Fraud. The transaction details, prediction result, and risk score are then stored in the SQLite database, and the dashboard and transaction history are updated. If the transaction is classified as Medium Risk or High Risk, the system generates an alert and sends an email notification to the registered user. This sequence ensures secure transaction processing, intelligent fraud detection, proper record management, and timely user notification.

10.5 Use Case Diagram

USE CASE DIAGRAM – CREDIT CARD FRAUD DETECTION AND TRANSACTION MONITORING SYSTEM



User (Primary Actor)

The User is the main actor who interacts with the system.

The user can perform the following operations:

- Register
- Login
- Manage Profile
- Add Bank Account Details
- Perform Transaction Analysis
- View Dashboard
- View Transaction History
- View Alerts
- Generate Reports
- Logout

Register:

The user creates a new account by entering personal and banking details.

During registration:

- User details are validated.
- Information is stored in the database.

Login:

The user enters their email and password.

The system:

- Validates the credentials.
- Authenticates the user.
- Allows access to the dashboard.

Manage Profile:

The user can update personal information and banking details whenever required.

Add Bank Account Details:

The user provides:

- Bank Name
- Account Number
- Card Number
- Current Balance
- Daily Transaction Limit
- Maximum Transaction Amount

These details are stored in the database and are later used by the Rule Engine during transaction validation.

Perform Transaction Analysis:

This is the core functionality of the project.

The user enters:

- Transaction Amount
- Transaction Hour
- Merchant Category
- Transaction Place
- Failed PIN Attempts
- Transactions in Last Hour
- Account Age

These transaction details are analyzed by the system.

Rule Engine Validation:

Before Machine Learning starts, the Rule Engine checks predefined banking rules.

The Rule Engine performs four validations:

Balance Check:

Checks whether:

Transaction Amount > Available Balance

If true,
the transaction is blocked.

Daily Limit Check:

Checks whether:

Today's transactions exceed the user's daily limit.

If yes,
the transaction is blocked.

Maximum Amount Check:

Checks whether:

Transaction Amount exceeds the user's maximum transaction amount.

If yes,
the transaction is blocked.

Failed PIN Attempts:

Checks the number of incorrect PIN attempts.

If the limit is exceeded,
the transaction is marked as High Risk.

ML Model Prediction:

If the transaction passes the Rule Engine,
it is sent to the **Random Forest Machine Learning model**.

The model predicts:

- Fraud Probability
- Risk Score

Calculate Risk Score:

The fraud probability is converted into a percentage.

Classify Risk Level:

Based on the risk score:

- **0–40%** → Safe
- **41–80%** → Medium Risk
- **81–100%** → High Risk / Fraud

Generate Decision:

The system decides whether to:

- Approve Transaction
- Allow with Monitoring
- Block Transaction

depending on the calculated risk score.

Store Transaction:

Every transaction is stored in the database along with:

- Transaction Details
- Prediction Result
- Risk Score
- Status
- Alert Information

Trigger Alerts:

If the transaction is classified as:

- Medium Risk
- High Risk

the system generates an alert.

The alert is stored in the Alerts module.

Send Email Notification:

Whenever:

- Medium Risk

- High Risk
- Rule Violation

occurs,

the system automatically sends an email notification to the registered user.

View Dashboard:

The user can monitor:

- Total Transactions
- Safe Transactions
- Fraud Transactions
- Blocked Transactions
- Current Balance
- Pie Chart
- Bar Chart
- Recent Transactions

View Transaction History:

Displays all previously analyzed transactions along with:

- Date
- Amount
- Risk Score
- Status

View Alerts:

Displays:

- Fraud Alerts
- Medium Risk Alerts
- Blocked Transactions

Generate Reports:

The user can generate:

- PDF Reports
- Excel Reports

These reports summarize transaction history and fraud detection results.

Logout:

The user safely exits the application, ending the session.

Database (Secondary Actor):

The Database stores:

- User Details
- Bank Information
- Transaction Records
- Fraud Alerts
- Risk Scores
- Reports

Email Service (Secondary Actor):

The **Email Service** sends notifications whenever:

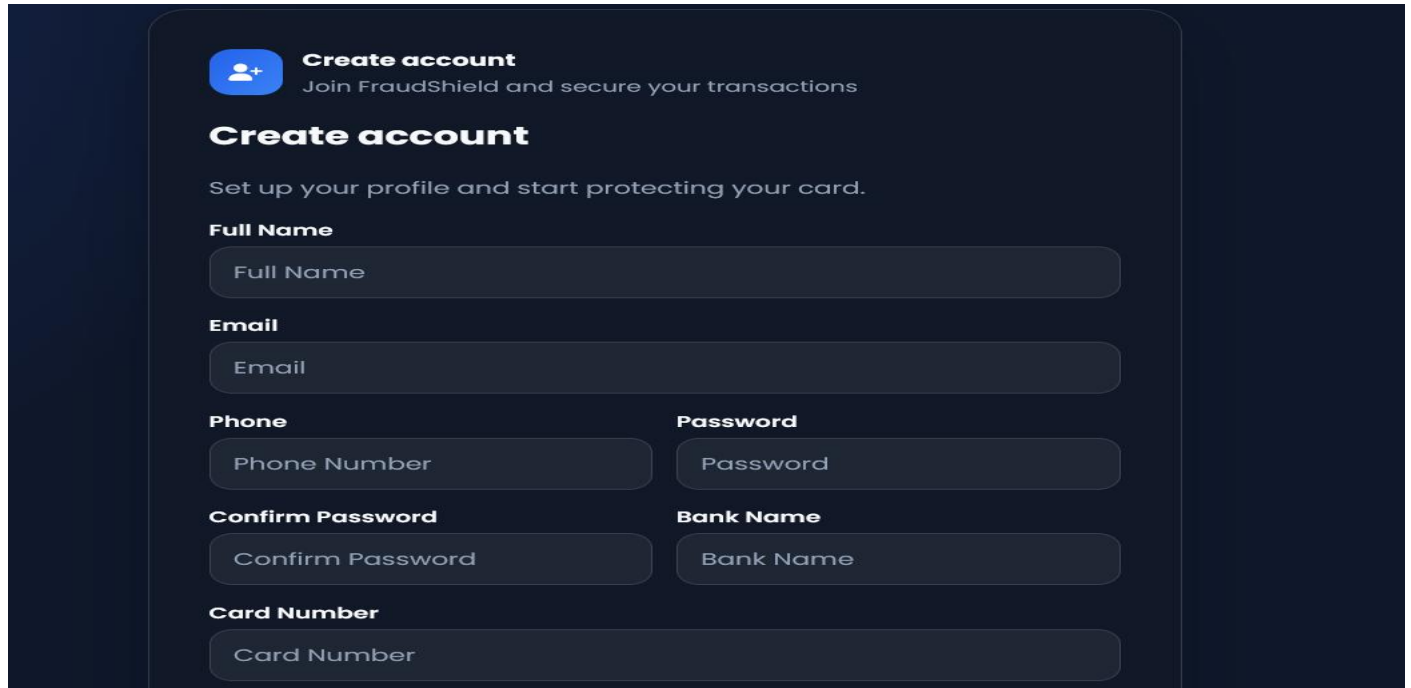
- A Medium Risk transaction is detected.
- A High Risk/Fraud transaction is detected.
- A Rule Engine validation fails (such as exceeding the daily transaction limit or maximum transaction amount).

11.Result and Discussion

Registration Module:

The registration module allows new users to create an account by entering personal and banking information.

The module successfully validates user input and stores the information securely in the database.



Create account
Join FraudShield and secure your transactions

Create account

Set up your profile and start protecting your card.

Full Name

Email

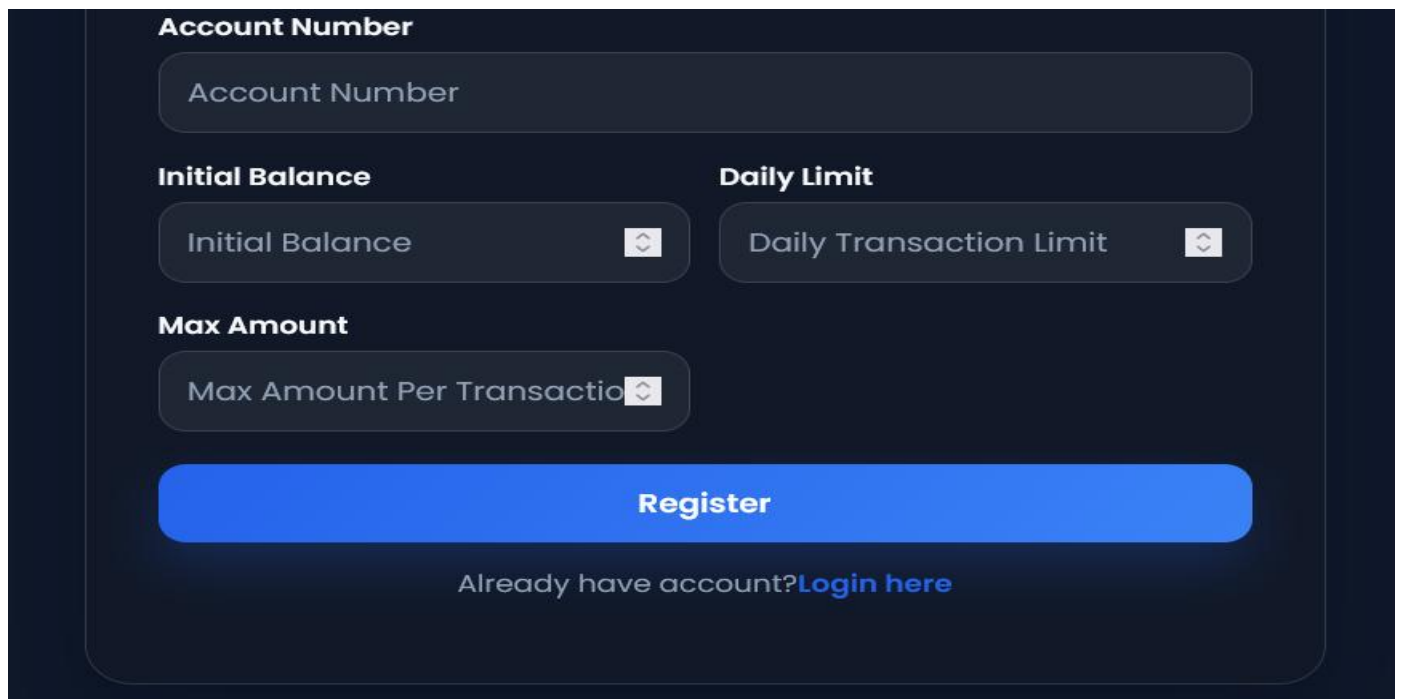
Phone

Password

Confirm Password

Bank Name

Card Number



Account Number

Initial Balance

Daily Limit

Max Amount

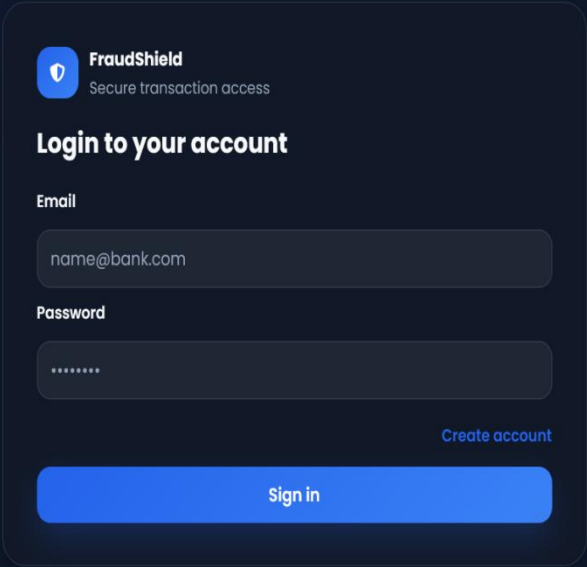
Register

Already have account? [Login here](#)

Login Module:

The authentication module verifies user credentials before granting access to the dashboard.

Only registered users can log in.



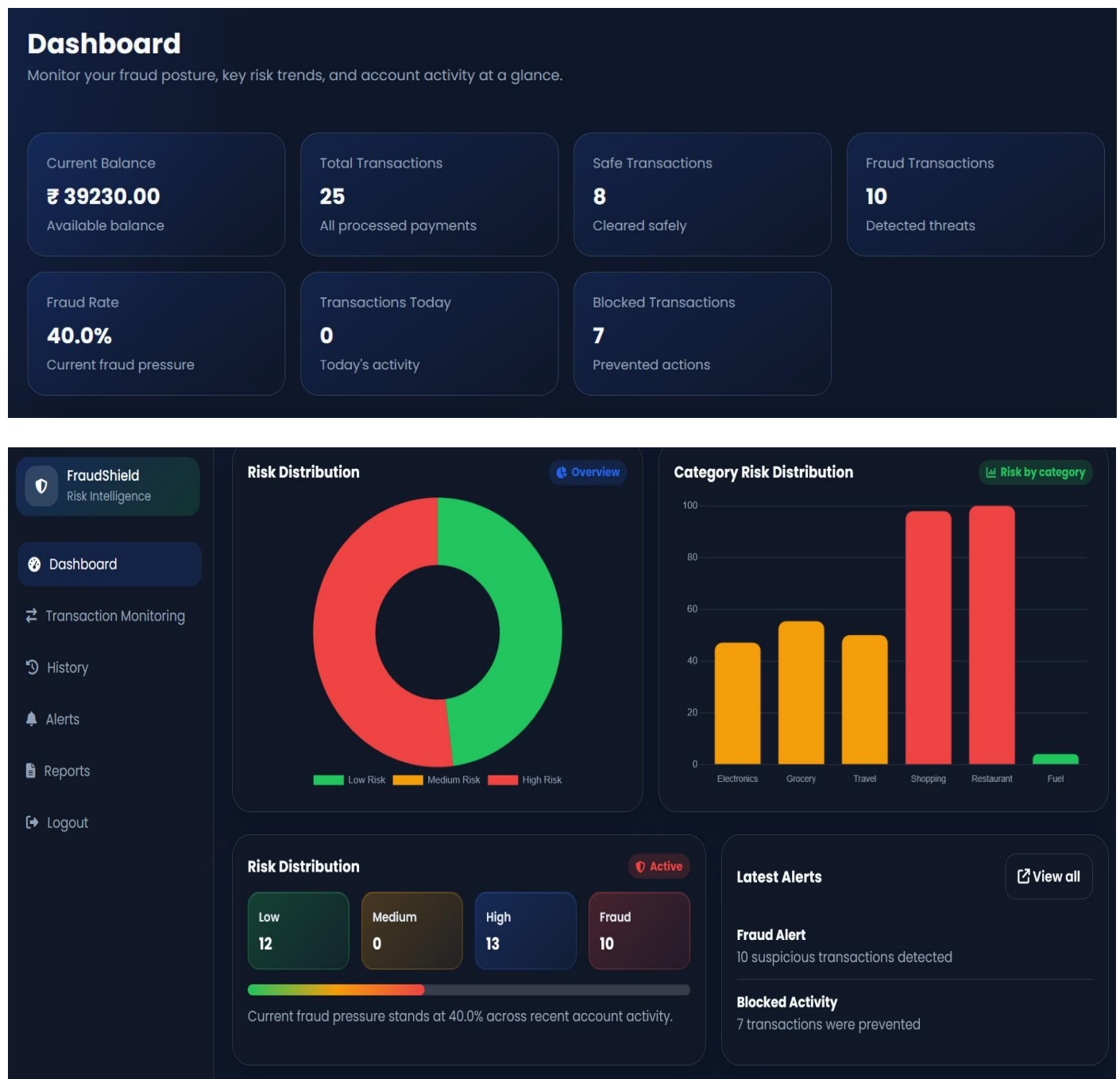
The image shows a login form for 'FraudShield' on a dark blue background. The form is a light blue rounded rectangle. At the top left is the 'FraudShield' logo, a blue shield with a white 'F', followed by the text 'FraudShield' and 'Secure transaction access'. Below this is the heading 'Login to your account'. There are two input fields: 'Email' with the placeholder 'name@bank.com' and 'Password' with a masked password '*****'. To the right of the password field is a link 'Create account'. At the bottom is a large blue button labeled 'Sign in'.

Dashboard Result:

The dashboard successfully displays:

- Total Transactions
- Fraud Transactions
- Safe Transactions
- Current Balance
- Pie Chart
- Bar Chart
- Recent Transactions

The graphical visualization helps users understand transaction statistics efficiently.



Transaction Monitoring :

Users enter transaction details through the transaction analysis form.

The system successfully accepts transaction information and forwards it for fraud analysis.

The Rule Engine successfully validates:

- Account Balance
- Daily Transaction Limit
- Maximum Transaction Amount
- Failed PIN Attempts

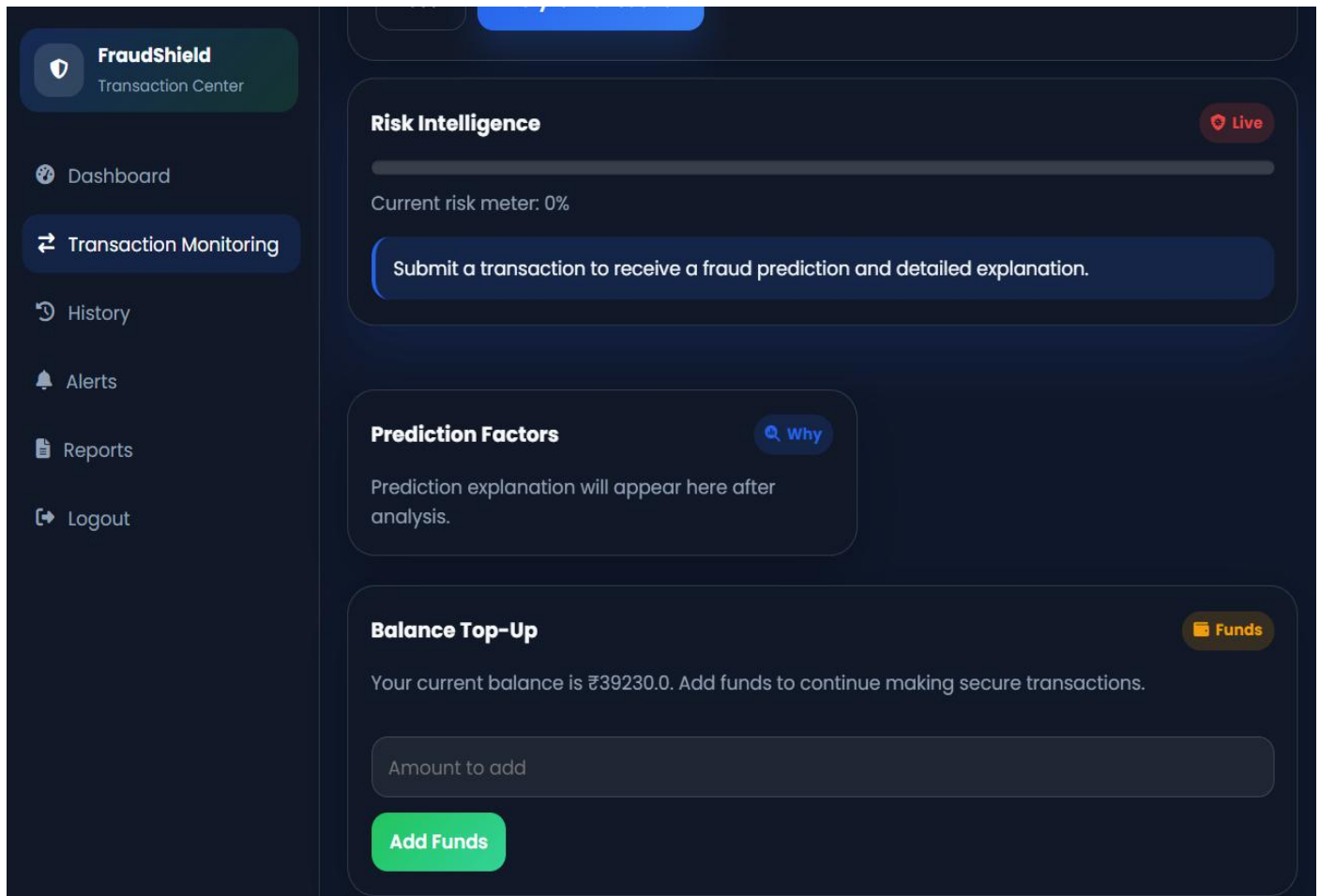
The screenshot displays the 'FraudShield Transaction Center' interface. On the left is a dark sidebar with navigation links: 'Dashboard', 'Transaction Monitoring' (highlighted), 'History', 'Alerts', 'Reports', and 'Logout'. The main content area is titled 'Analyze Transaction' and includes an 'AI Review' button. It features several input fields and dropdown menus for transaction analysis: 'Current Balance' (₹39230.00), 'Daily Limit' (3), 'Used Today' (0 / 3), 'Transaction Amount' (Amount), 'Transaction Hour' (12 AM), 'Transactions Last Hour' (Count), 'Failed PIN Attempts' (Attempts), 'Account Age Days' (Days), 'Merchant Category' (Electronics), 'Country Risk' (Low Risk Country), 'Card Present' (Card Present), and 'Transaction Type' (Online). At the bottom are 'Reset' and 'Analyze Transaction' buttons.

Field	Value
Current Balance	₹39230.00
Daily Limit	3
Used Today	0 / 3
Transaction Amount	Amount
Transaction Hour	12 AM
Transactions Last Hour	Count
Failed PIN Attempts	Attempts
Account Age Days	Days
Merchant Category	Electronics
Country Risk	Low Risk Country
Card Present	Card Present
Transaction Type	Online

Transactions violating these rules are blocked or marked as suspicious.

The Random Forest model successfully predicts fraudulent transactions.

The model calculates the fraud probability and generates a risk score.



History Module:

The History Module stores and displays all processed transactions. Users can view their previous transaction records along with important details.

The history displays:

- Date
- Amount
- Prediction
- Risk Score
- Status
- Balance After

FraudShield

Transaction History

Dashboard

Transaction Monitoring

History

Alerts

Reports

Logout

History

Review your recent transaction outcomes and past fraud assessments in one place.

Activity History

DATE	AMOUNT	PREDICTION	RISK SCORE	STATUS	BALANCE AFTER
2026-07-09	₹ 1500.0	FRAUD	100.0%	FRAUD	₹ 39230.0
2026-07-09	₹ 875.0	FRAUD	100.0%	FRAUD	₹ 39230.0
2026-07-07	₹ 10.0	SAFE	1.0%	BLOCKED	₹ 39230.0
2026-07-07	₹ 10.0	SAFE	1.0%	APPROVED	₹ 39230.0
2026-07-07	₹ 10.0	SAFE	1.0%	APPROVED	₹ 39240.0
2026-07-07	₹ 1000.0	FRAUD	95.0%	FRAUD	₹ 39250.0
2026-07-07	₹ 1000.0	FRAUD	95.0%	FRAUD	₹ 39250.0
2026-07-04	₹ 1000.0	SAFE	0.0%	APPROVED	₹ 39250.0

Alert Module:

Fraudulent transactions are automatically stored in the Alerts page.

The alerts display:

- Fraud Type
- Risk Score
- Alert Date
- Alert Reason

The screenshot displays the 'FraudShield Alert Center' interface. On the left is a dark sidebar with navigation links: 'Dashboard', 'Transaction Monitoring', 'History', 'Alerts' (highlighted), 'Reports', and 'Logout'. The main content area is titled 'Alerts' with the subtitle 'View fraud and security alerts'. Below the title are three filter buttons: 'All Alerts' (active), 'Fraud', and 'Blocked'. The alerts are listed in a table-like format with three rows:

Alert Type	Reason	Risk Score	Date
Fraud Detected	Machine learning fraud detection triggered	100.0%	2026-07-09
Fraud Detected	Machine learning fraud detection triggered	100.0%	2026-07-09
BLOCKED Transaction	Daily Transaction Limit Exceeded	1.0%	2026-07-07

Reports Module:

The system successfully generates:

- PDF Reports
- Excel Reports

The reports include transaction details, fraud prediction, and risk score.

FraudShield
Reporting Suite

Dashboard

Transaction Monitoring

History

Alerts

Reports

Logout

Reports

Filter, review, and export your transaction activity with clear risk summaries.

Report

Start Date

dd / mm / yyyy

End Date

dd / mm / yyyy

Status

All Status

Risk Level

All Risk Levels

Apply Filters

Download PDF

Download Excel

Report Range

Beginning → Now

Total Transactions

25

Fraud Transactions

10

Blocked Transactions

8

Safe Transactions

7

FraudShield
Reporting Suite

Dashboard

Transaction Monitoring

History

Alerts

Reports

Logout

Transaction Summary

Export-ready

DATE	AMOUNT	PREDICTION	RISK SCORE	STATUS	BALANCE AFTER
2026-07-09	₹ 1500.0	FRAUD	100.0%	FRAUD	₹ 39230.0
2026-07-09	₹ 875.0	FRAUD	100.0%	FRAUD	₹ 39230.0
2026-07-07	₹ 10.0	SAFE	1.0%	BLOCKED	₹ 39230.0
2026-07-07	₹ 10.0	SAFE	1.0%	APPROVED	₹ 39230.0
2026-07-07	₹ 10.0	SAFE	1.0%	APPROVED	₹ 39240.0
2026-07-07	₹ 1000.0	FRAUD	95.0%	FRAUD	₹ 39250.0
2026-07-07	₹ 1000.0	FRAUD	95.0%	FRAUD	₹ 39250.0
2026-07-04	₹ 1000.0	SAFE	0.0%	APPROVED	₹ 39250.0
2026-06-17	₹ 5000.0	FRAUD	98.0%	FRAUD	₹ 29250.0
2026-06-17	₹ 1000.0	SAFE	0.0%	APPROVED	₹ 29250.0

PDF Report Result (example):

Credit Card Fraud Report

Generated: 2026-07-07 11:36

User	Shivam GUpta
Total Transactions	29
Fraud Transactions	4
Blocked Transactions	11
Current Balance	₹41712.03

Transactions

Date	Amount	Prediction	Risk Score	Status	Balance After
2026-06-15 09:21:08	₹10000.0	FRAUD	98.0	FRAUD	₹22212.03
2026-06-15 08:05:58	₹10002.0	FRAUD	96.0	FRAUD	₹22212.03
2026-06-15 07:37:03	₹500.0	SAFE	0.0	APPROVED	₹22212.03
2026-06-13 11:03:19	₹7000.0	SAFE	7.0	APPROVED	₹2712.03
2026-06-13 10:45:49	₹288.0	SAFE	0.0	APPROVED	₹9712.03
2026-06-13 10:28:28	₹10000.0	SAFE	7.0	APPROVED	₹10000.03
2026-06-13 10:21:23	₹18785.0	FRAUD	91.0	FRAUD	₹20000.03
2026-06-11 12:15:21	₹1.0	SAFE	2.0	BLOCKED	₹20000.03
2026-06-11 12:13:44	₹10.0	SAFE	3.0	BLOCKED	₹20000.03
2026-06-11 11:06:39	₹100000.0	HIGH_RISK	99.0	BLOCKED	₹10000.03
2026-06-11 10:59:30	₹10000.0	HIGH_RISK	97.0	BLOCKED	₹10000.03
2026-06-11 10:55:37	₹10000.0	HIGH_RISK	100.0	BLOCKED	₹10000.03
2026-06-11 10:51:01	₹10000.0	SAFE	7.0	BLOCKED	₹10000.03
2026-06-11 10:47:20	₹10000.0	SAFE	7.0	BLOCKED	₹10000.03
2026-06-11 10:36:26	₹10000.0	SAFE	7.0	BLOCKED	₹10000.03
2026-06-11 06:59:03	₹10000.0	SAFE	93.0	BLOCKED	₹10000.03
2026-06-11 06:53:15	₹10000.0	SAFE	92.0	BLOCKED	₹10000.03
2026-06-11 06:52:31	₹9999.97	SAFE	6.0	APPROVED	₹10000.03
2026-06-11 06:51:06	₹10000.0	SAFE	6.0	APPROVED	₹10000.0
2026-06-11 06:50:39	₹10000.0	SAFE	4.0	APPROVED	₹20000.0
2026-06-11 06:49:39	₹10000.0	SAFE	7.0	APPROVED	₹30000.0
2026-06-10 20:52:58	₹20000.0	SAFE	5.0	APPROVED	₹0.0

Excel Report Result(example):

	A	B	C	D	E	F	G	H
1	ID	Date	Amount	Prediction	Risk Score	Status	Balance After	
2	59	2026-06-15 07:37:03	500	SAFE	0	APPROVED	22212.03	
3	58	2026-06-13 11:03:19	7000	SAFE	7	APPROVED	2712.03	
4	57	2026-06-13 10:45:49	288	SAFE	0	APPROVED	9712.03	
5	56	2026-06-13 10:28:28	10000	SAFE	7	APPROVED	10000.03	
6	55	2026-06-13 10:21:23	18785	FRAUD	91	FRAUD	20000.03	
7	49	2026-06-11 12:15:21	1	SAFE	2	BLOCKED	20000.03	
8	48	2026-06-11 12:13:44	10	SAFE	3	BLOCKED	20000.03	
9	46	2026-06-11 11:06:39	100000	HIGH_RISK	99	BLOCKED	10000.03	
10	45	2026-06-11 10:59:30	10000	HIGH_RISK	97	BLOCKED	10000.03	
11	44	2026-06-11 10:55:37	10000	HIGH_RISK	100	BLOCKED	10000.03	
12	43	2026-06-11 10:51:01	10000	SAFE	7	BLOCKED	10000.03	
13	42	2026-06-11 10:47:20	10000	SAFE	7	BLOCKED	10000.03	
14	41	2026-06-11 10:36:26	10000	SAFE	7	BLOCKED	10000.03	
15	36	2026-06-11 06:59:03	10000	SAFE	93	BLOCKED	10000.03	
16	35	2026-06-11 06:53:15	10000	SAFE	92	BLOCKED	10000.03	
17	34	2026-06-11 06:52:31	9999.97	SAFE	6	APPROVED	10000.03	
18	33	2026-06-11 06:51:06	10000	SAFE	6	APPROVED	10000	
19	32	2026-06-11 06:50:39	10000	SAFE	4	APPROVED	20000	
20	31	2026-06-11 06:49:39	10000	SAFE	7	APPROVED	30000	
21	22	2026-06-10 20:52:58	20000	SAFE	5	APPROVED	0	
22	21	2026-06-10 20:52:06	19800	SAFE	10	APPROVED	20000	
23	20	2026-06-10 20:51:34	20000	SAFE	0	APPROVED	39800	
24	19	2026-06-10 20:51:00	20000	SAFE	6	APPROVED	59800	
25	18	2026-06-10 20:49:47	21000	FRAUD	99	BLOCKED	79800	
26	17	2026-06-10 20:48:49	20000	SAFE	0	APPROVED	79800	
27	16	2026-06-10 20:43:18	70000	SAFE	7	PENDING_OTP	99800	
28	15	2026-06-10 20:41:50	200	SAFE	0	APPROVED	99800	

12.Future Scope

Although the proposed system performs fraud detection and transaction monitoring efficiently, several enhancements can further improve its capabilities.

The future scope includes:

1. Integration with Plaid Sandbox API for automatic bank account linking.
2. Support for live banking APIs to analyze real banking transactions.
3. Real-time transaction monitoring using webhooks.
4. OTP-based verification for medium-risk transactions.
5. SMS notifications for suspicious transactions.
6. Push notifications through a mobile application.
7. Integration of advanced Machine Learning algorithms such as XGBoost, LightGBM, and Deep Learning models.
8. Behavioral analysis based on user spending patterns.
9. Device fingerprinting to detect unauthorized devices.
10. Cloud deployment using AWS, Microsoft Azure, or Google Cloud Platform.
11. Multi-bank account support.
12. Role-based administration for banks and fraud analysts.
13. AI-powered fraud explanation using Explainable AI (XAI).
14. Multi-language support.
15. Mobile application development using Flutter or React Native.

13.Conclusion

The Credit Card Fraud Detection and Transaction Monitoring System Using Machine Learning successfully demonstrates the application of Machine Learning and Rule-Based Security in improving financial transaction security. The system provides secure user authentication, transaction analysis, fraud prediction, risk score generation, dashboard visualization, alert management, report generation, and email notification within a single integrated web application.

The use of the Random Forest Classifier enables accurate classification of transactions into Safe, Medium Risk, and High Risk categories based on transaction characteristics. The Rule-Based Security Engine further strengthens fraud prevention by validating banking rules before transaction processing.

The dashboard, graphical charts, transaction history, and reporting features provide users with a comprehensive view of their financial activities, while automated email notifications improve awareness of suspicious transactions.

Overall, the project achieves its primary objective of developing an intelligent, user-friendly, and scalable fraud detection system. The proposed architecture serves as a strong foundation for future integration with live banking APIs, real-time fraud detection, cloud deployment, and advanced security mechanisms, making it suitable for future financial technology applications.

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